Challenges in Perfume Recommender Systems: Navigating Subjectivity, Context and Sensory Data

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Abstract

Compared to other recommender systems domains, perfume recommendation proves to be highly personalized and more challenging due to the very subjective factors and complex mixture of involved senses. Individual perfume preferences are influenced by subtle elements such as emotional associations, personal memories, and unique biochemistry, making it difficult for users to clearly express their olfactory preferences. This paper provides an insight of significant challenges in perfume recommendations planned to be addressed in the context of my ongoing PhD project. By exploring these areas, I aim to make a meaningful contribution to the ongoing development of perfume recommender systems.

CCS Concepts

• **Information systems → Collaborative and social computing systems and tools**;

Keywords

Perfume Recommender System, Perfume Context, Perfume Emotional Perception

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1 Introduction

In e-commerce, personalized product recommendations are crucial for user satisfaction and increased sales.

Perfume is one of the most frequently used cosmetic products [5]. Compared to other domains, perfume recommendation proves to be highly personalized and more challenging due to the very subjective factors and complex mixture of senses: the primary sense of olfaction directly or indirectly stimulates several other senses: gustation, vision, tactile, auditory, memory and emotion. Certain authors even disregard the applicability of perfumes recommender systems, considering they could never replace a personal perfume advisor due to the very personal nature of a perfume [23].

Several perfume recommender systems create clusters of perfumes based on their similar notes [8], [15], [24], while only a smaller amount focus on users perception through sentiment or emotion analysis to appreciate the mood, feeling, personality of a perfume [13], [17]. Specialist perfumers recommend to not rely solely on fragrance notes when selecting a perfume, since the same note can be created using different compounds which may significantly change its actual scent.

Considering today's potential of infinite number of perfumes existing on the market, it is needed to explore the factors which facilitate the preference towards a perfume [3], considering that recommending a perfume is not only about matching notes, but also understanding individuals unique sensory preferences. The perfume transcends the typical definition of a consumer good, aligning more closely with an experience good [6], as its perception is deeply personal and inherently subjective.

Selecting the 'right' perfume goes beyond aligning with a list of notes - it involves evoking personal emotions, memories and expressions of identity. Therefore, effective perfume recommender systems must extend beyond basic feature matching, i.e. integrating guided discovery, scent literacy, and interactive preference refinement to genuinely assist individuals in their fragrance exploration.

Developing effective perfume recommender systems holds considerable importance from multiple perspectives. Firstly, by providing personalized and relevant suggestions, the system aids users in navigating the complex and often confusing landscape of scents, facilitating the discovery of perfumes they will enjoy, including niche or unexpected fragrances. Secondly, from commercial impact perspective, recommender systems reduce user search effort and boost e-commerce sales. Thus, accurate suggestions can enhance sales and strengthen customer loyalty. Thirdly, creating robust perfume recommender systems marks a key step in advancing personalization within a highly subjective and nuanced domain. By aligning suggestions with individual scent preferences, emotional responses and contextual factors (e.g. occasion, season), such systems push the limits of current recommendation technologies.

This research addresses the inherent challenges of perfume recommendation by providing insight into the key aspects that complicate the process. Following this, we present our prior work in this domain, along with current progress and future directions.

2 Research Questions and Directions

RQ1: Surrounding Context for wearing a perfume - might a scent be enjoyed in one setting and disliked in another?

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Daily and contextual aspects shall be taken into consideration when providing recommendations. Context can be represented by an occasion, the season when a person would wears the perfume or even time of day, for example if the person plans to wear the perfume during the day (office, regular daily activities) or during the night (events, parties).

Consumers tend to choose fragrances based on ocasion and social norms [7]. Having diplomatic olfactory and wearing occasion-suitable perfumes shapes others opinion towards an individual and influences social judgments [10] or possible harassments [12].

Perfume advertisements use different types of semiotic modes - verbal, visual, olfactory - to convey meaning [19]. Similar and poetic words, repeated or related images are blended to create a description of the smell and feeling, together with the setting or atmosphere that best complements a fragrance.

Capturing the highly context-sensitive nature of perfume represents a significant challenge in the development of effective recommender systems. Individual scent choices are not only influenced by personal taste and biochemistry, but also vary depending on contextual factors such as season, occasions, cultural norms. These dynamic influences make it difficult to build static models of preference, as the same individual may prefer entirely different fragrances in different contexts. Addressing this complexity requires context-aware modeling techniques that can adapt to changing user preferences and interpret subtle cues from both explicit user feedback and subtle interaction.

RQ2: Individual Olfactory Perception - Analysis of individual experiences, memories and emotional associations with scents.

Olfaction and emotion are closely related [25]. Perfumes have the ability to evoke emotions, memories, create personal connections [7], even improve the quality of life and prevent sickness [9]. In addition to the emotional impact of smelling a perfume, psychological and emotional benefits emerge from other perfume-oriented interactions, such as creating DIY (Do It Yourself) perfumes[16].

Analyzing emotional responses evoked by odors sensitivity, paper [21] observes clear differences in behavioral and physiological indicators (respiration, skin conductance, facial electromyography and heart rate) between odors (food, cosmetics, animal) and fragrances (essential oils or fine perfumes). Moreover, was identified that unpleasant olfactory stimuli are more intense than pleasant stimuli.

Paper [11] uses several types of emotional measurements to reflect the emotional reaction to perfume across Germans and Chinese. The study reveals different preferences of odor between the two nations (due to the cultural backgrounds, daily life and habits), and also different tendency to express emotions as response to odors: Germans express more with facial expressions, while Chinese express more with the body and skin conductance.

The scent experience is also influenced by visual aspects, such as the packaging which can lead to different emotional responses [4]. An aesthetic and quality package has a higher purchase intention [14]. On the other hand, after smelling a perfume, certain visual correlations can be done, such as color-perfume pairing [22], gender-perfume pairing[26].

Understanding the interplay between scent and individual perception opens new area for research in affective computing and experiential recommendation. To enhance personalization, perfume recommendation models must explore techniques for scent-associated memories, emotion-driven tagging, or even physiological responses to scents.

RQ3: Impact of Perfumer or Perfume Houses in the scent creation and complexity

The perfume reflects the training, philosophy, cultural background and individual preferences of the perfumer.

For example, certain perfumers like Jean-Claude Ellena or Jacques Cavallier are known for simple and clean perfumes, as they often emphasize clarity and minimalism, leading to a beautiful rise of the individual notes.

Other perfumers, like Dominique Ropion or Francis Kurkdjian, prefer to create multi-layered, rich perfumes, which balance complex compositions and evolving structures.

The third perfumers category balances the simplicity and the maximalism, refering to perfumers who shift their creation styles depending on the brand and the requirements, e.g. Pierre Bourdon or Olivier Polge.

Most major perfume houses have a specific olfactory DNA, which refers to a set of stylistic, conceptual or ingredient choices that define the brand. These olfactory DNAs help ensure a consistent brand identify, regardless of the different perfumers who are involved, and are often reflecting the brand history or artistic direction.

Moreover, people are more attracted towards branded or luxury perfumes, and are inclined to rate the perfumes as more pleasant when they are presented in a luxurious context [2].

Incorporating information about the perfumer and perfume house into recommendation models could provide an additional layer of personalization and interpretability. Knowing a user's preference for the style of a particular perfumer or perfume house could guide recommendations beyond note-based similarity, capturing subtle yet meaningful factors that influence user satisfaction. Moreover, this perspective acknowledges that scent preference is not only about olfactory composition but also about story-telling, artistry, and brand affiliation.

RQ4: Perfume Representation - Fuzzy Vector Representation of a Perfume

Perfume features are intrinsically vague and perceived differently by individuals. Traditional crisp representation often fail to capture these subtleties. In contrast, fuzzy set theory provides the possibility to better model perfume features and improve the sensory analysis.

Sensory perceptions are highly subjective, allowing a wide range of interpretation. Using fuzzy vectors, a feature can simultaneously belong to multiple categories to varying degrees, thus reflecting the natural uncertainty and ambiguity present in human perceptual expression.

For example, a perfume can be modeled as a vector of fuzzy features, where each feature corresponds to a sensory or performance dimension such as Fragrance Family (FF), Longevity (LG), Sillage (SL).

Fragrance Family refers to a refined classification of perfumes into distinctive olfactory groups. The number and nature of these groups depend on the chosen classification approach. Most widely recognized olfactory groups are Floral, Oriental, Woody and Fresh.

Longevity measures the duration of time a perfume remains perceptible on the skin after application, usually described using qualitative terms such as short, moderate, long-lasting.

Sillage describes the degree in which a fragrance projects into the air and leaves a perceptible scent trail as the wearer moves. The sillage or strength of the fragrance is often classified as light or intimate, moderate, strong.

$$P = \{FF, LG, SL\}, where$$

where:

- FF is a fuzzy set with support set {Floral, Oriental, Woody, Fresh}
- LG is a fuzzy set with support set {Short, Moderate, Long}
- SL is a fuzzy set with support set {Light, Moderate, Strong}

The membership degrees for the fuzzy sets can be defined through various methods: expert-defined, data-driven, collaborative. In our research, we plan to use Collaborative Fuzzification, to capture the collective perception from perfume dedicated social media.

The fuzzy perfume representation enables a more effective comparison between perfumes using fuzzy similarity measures and enables fuzzy clustering of perfume profiles, ultimately improving recommendation systems for providing more accurate and personalized recommendations.

Natural Language Processing (NLP) techniques and Large Language Models (LLMs) can be used to extract nuanced features and user preferences from textual descriptions and reviews, enabling the development of a sophisticated computational model for perfume recommendation.

Central goal of this RQ is a novel and rich feature representation of perfumes that captures their multifaceted nature — from chemical composition to olfactory character and emotional resonance. A key challenge lies in effective data fusion, as the system must integrate heterogeneous sources such as user reviews, scent descriptors, emotional responses, and contextual metadata.

RQ5: Recommendations Evaluation - Appropriate methodologies for sensory-rich domain

In evaluating the effectiveness of perfume recommender systems, it is essential to go beyond traditional accuracy metrics and incorporate beyond-accuracy measures that reflect the quality and user-centric value of the recommendations [1].

Novelty refers to the ability of the system to suggest less popular or previously unexplored items, often using item popularity scores or user interaction histories.

Diversity evaluates the variety within a recommendation list. In the the context of perfumes, diversity measures differences in olfactory families or fragrance notes.

Serendipity captures the capacity of the system to generate unexpectedly relevant recommendations, commonly defined as the product of unexpectedness and relevance.

Coverage measures the proportion of the overall perfume catalog that the system is able to recommend, indicating its ability to utilize the full breadth of available items.

Together, these metrics provide a more holistic evaluation of system performance in real-world scenarios, capturing both objective accuracy and subjective nuances of user perception in a sensory-rich domain.

Offline metrics provide scalable evaluation but may not fully capture the user experience for sensory products.

Online evaluation through user studies is essential for assessing the real-world effectiveness and user experience of perfume recommender systems. Various methodologies can be employed to capture meaningful feedback, including task-based evaluations, system comparison with baseline models in live or simulated environments, semi-structured interviews.

In these evaluations, a wide range of user-centric metrics are considered, including perceived accuracy and relevance, satisfaction, trust, perceived novelty, diversity, and serendipity, as well as the usefulness and clarity of recommendations.

3 Preliminary Results and Future Directions

In context of perfume recommmender systems, we have proposed a method for obtaining perfume recommendations starting from user lifestyle [18], which uses a questionnaire to get the user preferences regarding the contextual details of wearing a perfume (season, time of day, longevity, silage, gender), together with the Fragrance Family. The families which are included in our system are Floral, Oriental, Woody, Aromatic, Citric, Green, Fruity, Aquatic.

We evaluated the system by submitting a set of randomly generated questionnaires, to simulate a population of users filling the questionnaire and seeking for recommendations. The results show that almost 93% of the recommendations provided by the system are identified as relevant or valuable recommendations. A recommendation is considered relevant, if the respective perfume was voted as a pleasant scent by large majority of the fragrance community.

We plan to improve the system such that the questionnaire takes into consideration a more detailed perfume context description: formal event, casual event, work, dinner, wedding etc. This kind of detailed description can be collected as a textual field and processed using NLP techniques, and then the identified contextual details can be mapped with the perfumes details, e.g. based on the perfume notes [20].

Our current research is focused on investigating methods to represent perfumes and derive rich semantic representations. Understanding the individual experiences is a key factor to ensure qualitative and personalized perfume recommendations.

We represent perfumes as vectors of different characteristics, combining elements from sensory science, data science, natural language processing and fuzzy logic.

We plan to implement a fuzzy-based recommender system which uses fuzzy similarity measures to identify and match individual user preferences. We intend to incorporate contextual filters related to perfume context suitability (e.g. season, occasion) and employ rule-based fuzzy inference to enhance recommendation quality.

The recommender system aims to provide accurate, diverse, and serendipitous recommendations that are also explainable to the user.

To ensure a thorough evaluation of the recommender system, we will conduct rigorous offline experiments employing both established and newly adapted metrics. Complementing these with online user studies will enable us to assess real-world effectiveness and user satisfaction.

Exploring these research topics will enhance our understanding of how to design effective and trustworthy recommender systems tailored to complex sensory products.

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